

ROMIL V. SHAH

Software Engineer | Backend · AI/LLM · Security
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SUMMARY

Software engineer with 3+ years shipping production backends, ETL pipelines, and security tooling on AWS and Azure - Python and C# REST APIs, data pipelines at 8M+ rows/day, and detection automation on Splunk and MITRE ATT&CK. I run the Claude API in production - concurrent, orchestrated calls via Orkes Conductor with p50/p95 latency and token-cost instrumentation. I own systems end-to-end, from threat model to CI.

EXPERIENCE

Unique Design Inc., Secaucus, NJ - Software Developer

Apr 2025 - Jul 2026

- Designed and operated production ETL pipelines (SSIS, Sybase/PostgreSQL to AWS RDS) processing 8M+ rows daily, powering Tableau dashboards for 250+ users and feeding real-time SIEM incident data into analyst triage queues.
- Built Splunk correlation rules and automated triage workflows across customer environments, reducing false-positive rates and routing confirmed incidents to analysts within contractual SLA windows.
- Architected C# backend modules and Bash automation on UNIX/Windows to query SQL Server and PostgreSQL, cutting document and metadata retrieval latency by 60%.

Dianco Inc., New York City, NY - Software Engineer

Mar 2024 - Apr 2025

- Wrote Python services to parse and correlate firewall logs in real time, integrating with Splunk to surface actionable security incidents within 5 minutes of detection.
- Built SSIS ETL pipelines and SQL synchronization jobs to move PostgreSQL e-commerce data into SQL Server, eliminating 95% of sync errors and recovering 10 hours of manual work weekly.
- Deployed Azure IaaS and Front Door CDN, applied CIS Level 1 hardening across the production environment, and sustained 99.25% uptime.

iTechNuts Solutions, Ahmedabad, India - Software Engineer

Jan 2021 - Mar 2022

- Built and shipped backend logic and gameplay systems for educational mobile games across iOS and Android, collaborating with designers to meet App Store and Play Store release requirements.

PROJECTS

AI-Powered Log Analyzer (Python, Go, Claude API, Kafka, PyTorch, Docker, GitHub Actions) - Spring 2026 | github.com/Romil2112/log-analyzer

- Built a Python security CLI that detects brute-force attacks (T1110.001), port scans (T1046), and web scans (T1595.002) from SSH, Windows, and Apache/Nginx logs using dual ML scoring (Isolation Forest + PyTorch autoencoder). A labeled eval harness raised detection precision to 1.0, and pySigma exports to Splunk, Elastic, and Sentinel with full CIM/ECS/ASIM schema coverage.
- A multi-step Claude tool-use agent (4 tools, MAX_ROUNDS=5) queries RAG over MITRE ATT&CK STIX embeddings (fastembed + pgvector) to write context-aware summaries and Sigma rules on demand, orchestrated as an Orkes Conductor fork/join workflow with p50/p95 and token-cost instrumentation. Deployed on Kubernetes/Helm + Terraform/GCP with 372 pytest tests, a Go log-shipper across HTTP REST, gRPC, and Kafka transports, and a --gcs-bucket flag for GCS report upload.

SOC Dashboard (Python, Go, Flask, PostgreSQL, Kafka, Redis, gRPC, Chart.js) - Spring 2026 | github.com/Romil2112/SOC-Dashboard

- Built a full-stack SOC triage dashboard (REST ingestion from the Log Analyzer, severity-ranked queue, run provenance per alert) secured by Flask-Login/bcrypt RBAC, CSRF + HMAC-verified API keys, Fernet encryption at rest, and an atomic audit trail. A Go ingest microservice exposes dual REST (:8001) and gRPC (:9001) transports, with cross-language Fernet parity verified by a CI Go-Python roundtrip test.
- A Kafka consumer drives ingest, Redis pub/sub delivers real-time SSE alerts to the browser (EventSource primary, polling fallback), and pgvector scores each alert against its top-5 historical matches. SQL aggregate JOINS surface MTTR, per-severity SLA, and escalation KPIs 2x faster, with server-side pagination (LIMIT/OFFSET, idx_alerts_created_at) and 229 tests (142 pytest + 87 Go) with live PostgreSQL in CI.

Face Tracking System (Python, FastAPI, OpenCV, YOLO, Docker) - Spring 2025 | github.com/Romil2112/Face-Tracking-System

- Built a fault-tolerant real-time face detection pipeline using YOLOv8-nano (ultralytics) with a Haar Cascade fallback, running across CUDA, OpenCL, and CPU with circuit-breaker recovery, adaptive frame-skipping, and 5-frame temporal filtering.
- Packaged as a production FastAPI microservice with per-IP rate limiting, Prometheus /metrics, mediapipe EAR liveness detection for anti-spoofing, and Claude Haiku triage on low-confidence detections, with 224 pytest tests across Python 3.10-3.12.

EDUCATION

Stevens Institute of Technology, Hoboken, NJ

Jan 2022 - Dec 2023

MS Computer Science; Graduate Certificate in Cybersecurity

Marwadi University, Rajkot, India

Jun 2018 - May 2021

Bachelor of Computer Applications

SKILLS

Programming & Backend: Python, Go, C#/.NET, Java, C++, Bash, PowerShell; REST API design (Flask, FastAPI), gRPC

Data & Pipelines: SQL Server, PostgreSQL, MySQL, Sybase, ETL pipelines, SSIS, AWS RDS, Kafka, Redis, Elasticsearch, scikit-learn (Isolation Forest), PyTorch (autoencoder), NumPy

Cloud & Infra: AWS (EC2, RDS, S3, Lambda), GCP (Cloud Run, GKE, Cloud SQL, Cloud Storage), Azure IaaS, Azure AD / SSO (SAML), Kubernetes, Helm, Terraform, UNIX, Windows Server

DevOps & Tooling: Docker, GitHub Actions CI/CD, Prometheus, OpenTelemetry, Azure DevOps, Git, pytest

LLM & AI Tools: Generative AI, Large Language Models (LLMs), Claude Code (Anthropic), Claude API, prompt engineering, agentic workflows & orchestration, Orkes Conductor, concurrent batched LLM calls, token-cost/p50/p95 latency instrumentation, RAG, pgvector embeddings

Security Domain: Splunk/Elastic/Sentinel SIEM, Sigma detection-as-code, detection-rule versioning, precision/recall/F1 evaluation harness, firewall log analysis, MITRE ATT&CK, CIS Level 1 hardening, Incident Response; Flask-Login/bcrypt auth, role-based access control (RBAC), atomic audit trails, Fernet field-level encryption + HMAC integrity, PII/IP scrubbing & pseudonymization, data-retention controls

Visualization: Tableau, Chart.js, Visio, OpenCV, mediapipe, YOLO (ultralytics)